Lecture 2

Algorithmic fairness: guarantees, conflicts, and context

AI-assisted reconstruction from course slides, not a lecture transcript. Discussion reasoning is inferred; classroom responses were not supplied.

The lesson. Fairness criteria compare different populations and protect different interests. Choose the comparison explicitly, check its assumptions, and connect it to the decision's consequences. Equal rates do not establish individual fairness, appropriate labels, or a fair institutional process.

Which comparison is being made?

Let A denote group, Y the recorded binary outcome, R a score, and $\widehat{Y}$ the binary decision. Discrimination can affect allocation, quality of service, or representation. Excluding A from inputs leaves proxies and biased labels: the slides' school-based rule awards 80% of one group and 20% of another despite never reading the group label.

Criterion	What is equal across groups?
Demographic parity	$\mathbb{P}(\widehat{Y}=1 \mid A)$; independence: $\widehat{Y} \perp\!\!\!\perp A$.
Equal opportunity	TPR: $\mathbb{P}(\widehat{Y}=1 \mid Y=1, A)$.
Equalized odds	TPR and FPR; separation: $\widehat{Y} \perp\!\!\!\perp A \mid Y$.
Predictive parity	PPV: $\mathbb{P}(Y=1 \mid \widehat{Y}=1, A)$.
Sufficiency	$Y \perp\!\!\!\perp A \mid R$; binary decisions require equal PPV *and* NPV.
Group calibration	$\mathbb{P}(Y=1 \mid R=r, A=a)=r$ for every supported score and group.

Calibration overall can conceal group miscalibration: the slides' score 0.7 has outcome rates 0.8 and 0.6 in two groups, averaging to 0.7. Group calibration implies score sufficiency where groups share support. Thresholding a calibrated score need not preserve predictive parity: groups can have different mixtures of scores above the cutoff. Condition only on events with positive probability.

Two conflicts worth being able to derive

Write $\pi_a = \mathbb{P}(Y=1 \mid A=a)$ and $q_a = \mathbb{P}(\widehat{Y}=1 \mid A=a)$. Then

$$q_a = \mathrm{TPR}_a \pi_a + \mathrm{FPR}_a(1-\pi_a).$$

Under equalized odds, with common TPR u and FPR v, $q_a - q_b = (u-v)(\pi_a - \pi_b)$. Unequal base rates and $u \neq v$ prevent demographic parity. The assumptions matter: $u = v$ is an exception, and the binary-outcome argument should not be extended blindly to multiclass targets.

Bayes' rule gives the second conflict:

$$\mathrm{PPV}_a = \frac{\mathrm{TPR}_a \pi_a}{\mathrm{TPR}_a \pi_a + \mathrm{FPR}_a(1-\pi_a)}.$$

With $0 < \pi_a, \pi_b < 1$, unequal base rates and common *positive* TPR and FPR imply unequal PPV. Perfect prediction is an important exception. The COMPAS discussion uses this tension to explain why predictive parity and equal error rates can disagree, without deciding which criterion a court should prioritize.

Errors acquire consequences through policy. If a high-risk label changes a penalty from $t_{\min}$ to $t_{\max}$, the expected penalty for recorded $Y=0$ in group a is $t_{\min} + (t_{\max} - t_{\min})\mathrm{FPR}_a$. Using the same score to offer support changes the meaning of that false positive.

Individual fairness moves the hard choice into a metric

Dwork et al. model a randomized decision as $M(x) \in \Delta(O)$ and require

$$D_{\mathrm{TV}}(M(x), M(x')) \le d(x, x'), \qquad D_{\mathrm{TV}}(P, Q) = \tfrac{1}{2} \sum_{o} |P(o) - Q(o)|.$$

For binary decisions with award probabilities $p_x, p_{x'}$, total variation is $|p_x - p_{x'}|$. With $d(x, x') = 0.1$, probabilities $0.8, 0.3$ violate the constraint; $0.55, 0.45$ satisfy it. The guarantee concerns distributions over treatment, not equality of every realized draw.

Equal group award rates can still treat similar cross-group pairs differently. Conversely, individual fairness permits a group gap if groups differ under the chosen task metric. If S, T are distributions over individuals and μ_S, μ_T their induced outcome distributions, then

$$D_{\mathrm{TV}}(\mu_S, \mu_T) \le d_{\mathrm{EM}}(S, T), \qquad d_{\mathrm{EM}}(S, T) = \min_{\gamma \in \Pi(S,T)} \mathbb{E}_{\gamma}[d(x, x')].$$

Earthmover distance is the cheapest average task distance for transporting one group distribution to the other. Close groups force close outcome distributions. For the slides' groups at $\{.1, .3\}$ and $\{.6, .8\}$ with equal masses, $d_{\mathrm{EM}} = .5$: the fair rule $p_x = x$ attains gap .5, whereas $p_x = .5$ has gap zero. A bound permits a disparity; it does not require it. The metric's relevance, scale, and cross-group comparisons need justification.

Where interventions act, and what A1 adds

Preprocessing changes data, weights, labels, samples, or representations. Adversarial representation learning trades task information against the ability to infer group membership; synthetic-data methods also impose assumptions. **In-processing** adds constraints or penalties to training. **Post-processing** changes decisions, for example via thresholds or rejection options. Feasibility, access to group information, utility, and transfer to new settings depend on the method; the slides' comparison is illustrative.

A1's smooth penalty can use $(\overline{R}_{A=0} - \overline{R}_{A=1})^2$ alongside prediction loss. It supplies gradients, unlike hard thresholded decisions, but score-mean parity does not guarantee binary-decision parity. Evaluate the hard DP gap separately. Reweighted accuracy averages the two group accuracies equally; ordinary accuracy weights by group frequency. Compute evaluation rates on the full evaluation set.

The five abstraction traps explain what a metric omits

Framing: a narrow model boundary omits how outputs become decisions. **Portability:** social context changes what a method means, beyond a change in data distribution. **Formalism:** a mathematical condition captures only part of fairness. **Ripple effect:** deployment changes behaviour and institutions. **Solutionism:** beginning with a model excludes policy changes or not deploying. A criterion alone cannot establish legal compliance.

Midterm emphasis (inference). Highest priority: compute and interpret group rates, distinguish conditioning directions, and derive conflicts with their exceptions. Also practise the individual-fairness bound, intervention choices, and applying an abstraction trap to a concrete case. The reference sheet supplies the main group criteria, but not the individual-fairness or earthmover formulas.

Slide trail. L02.pdf, pp. 6–18 (group criteria), 19–30 (interventions/A1), 31–45 (Dwork et al.), 46–55 (Chouldechova), 56–65 (Selbst et al.); annotated diagrams consulted. A1: pp. 1–3. Page numbers refer to PDF pages, not the slides' continuous printed numbering.